

AI Mental Health Analysis Report
DS2_Task - Hypothesis testing setup
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Submitted By: Anurag Chawla
Data Science Intern

1. Introduction

This report presents a statistical analysis of mental health data from the Adult Psychiatric Morbidity Survey (APMS) 2014. The objective is to examine whether gender and age have a significant effect on the prevalence of Common Mental Disorders (CMD).

The survey captured responses from 7,546 adults aged 16 and over, sampled across all nine Government Office Regions of England using a robust stratified, multi-stage probability sampling design. Data collection took place during 2014 representing the most comprehensive and reliable profile of adult mental health in England available at that time.

Hypothesis testing techniques such as Chi-square tests and t-tests are used to validate the relationships between variables.

2. Objective

The primary aim of this analysis is to apply inferential statistical techniques to mental health prevalence data from the APMS 2014 survey, in order to identify and quantify meaningful demographic patterns in Common Mental Disorder (CMD) occurrence across the English adult population.

The specific objectives of this analysis are:

- To determine whether CMD prevalence differs significantly between males and females using chi-square association testing
- To examine whether CMD prevalence varies significantly across age groups (16–24 through to 75+) at a population level
- To compare mean CMD rates between male and female adults across all age bands using independent samples t-testing

3. Dataset Description

The data used in this analysis is sourced entirely from the **Adult Psychiatric Morbidity Survey (APMS) 2014**, published by NHS Digital. The survey represents the fourth in a series of nationally representative psychiatric morbidity surveys and remains the most comprehensive assessment of adult mental health prevalence in England to date.

3.1 Source File

File: apms-2014-ch-02-tabs.xls **Chapter:** Chapter 2 — Common Mental Disorders
Publisher: NHS Digital / Health and Social Care Information Centre (HSCIC) **Survey Year:** 2014 (fieldwork conducted 2014)

3.2 Specific Table Used

All hypothesis testing in this analysis is performed using data extracted from **Table 2.3 "Any CMD in past week, by age and sex"**. This table was selected because it provides:

- CMD prevalence rates measured using **diagnostic criteria** for six specific disorder types, aggregated into a single "Any CMD" indicator
- A complete breakdown by both **gender** (Male, Female) and **age group** simultaneously
- **Unweighted sample sizes** at the base of the table, which are essential for computing observed counts in chi-square tests and standard deviations in t-tests

3.3 Key Variables Extracted:

Variable	Description	Values
Age Group	Seven age bands covering full adult population	16–24, 25–34, 35–44, 45–54, 55–64, 65–74, 75+
Gender	Binary sex classification as recorded in survey	Male, Female
Any CMD Prevalence (%)	Percentage of adults meeting diagnostic criteria for at least one CMD in the past week	Continuous: 5.6% – 28.2%
Sample Size (N)	Unweighted count of survey respondents per subgroup	Male total: 3,058 · Female total: 4,488 · All: 7,546

Age	Male_CMD_%	Female_CMD_%	Male_N	Female_N
0 16-24	10.0	28.2	249	311
1 25-34	17.4	20.7	355	680
2 35-44	16.3	22.3	468	712
3 45-54	13.8	24.2	489	805
4 55-64	15.6	20.2	541	685
5 65-74	8.1	14.7	538	651
6 75+	5.6	11.0	418	644

4. Methodology

4.1 Data Preparation

- Converted CMD percentages into actual counts

	Age	Male_CMD_%	Female_CMD_%	Male_N	Female_N	Male_CMD_Count	Female_CMD_Count	Male_No_CMD	Female_No_CMD
0	16-24	10.0	28.2	249	311	25	88	224	223
1	25-34	17.4	20.7	355	680	62	141	293	539
2	35-44	16.3	22.3	468	712	76	159	392	553
3	45-54	13.8	24.2	489	805	67	195	422	610
4	55-64	15.6	20.2	541	685	84	138	457	547
5	65-74	8.1	14.7	538	651	44	96	494	555
6	75+	5.6	11.0	418	644	23	71	395	573

- Created contingency tables for hypothesis testing

4.2 Hypothesis Definitions

1. Gender vs CMD (Chi-square test)

- **H0 (Null Hypothesis):** No association between gender and CMD prevalence
- **H1 (Alternative Hypothesis):** There is an association

2. Age vs CMD (Chi-square test)

- **H0:** No association between age and CMD prevalence
- **H1:** There is an association

3. Male vs Female CMD Rates (t-test)

- **H0:** Mean CMD rate is equal for males and females
- **H1:** Mean CMD rate is different.

5. Results

The statistical analysis reveals strong evidence of relationships between demographic factors and CMD prevalence.

- **Gender vs CMD:**
The Chi-square test produced a p-value of **<0.001**, indicating a statistically significant association between gender and CMD prevalence.

This suggests that CMD rates differ meaningfully between males and females.

```
*** Chi-square (Gender vs CMD):
Chi2: 69.27513432944679
p-value: 8.564254470717197e-17
```

- **Age vs CMD:**
The Chi-square test yielded a p-value of **<0.001**, confirming a significant relationship between age and CMD prevalence.

CMD prevalence varies across different age groups.

```
Chi-square (Age vs CMD):  
Chi2: 100.53555570784262  
p-value: 1.9400175267719785e-19
```

- **Male vs Female CMD Rates:**

The t-test resulted in a p-value of **0.0158**, indicating a statistically significant difference in CMD rates between males and females.

```
... T-test (Male vs Female CMD %):  
t-stat: -2.8099130342890377  
p-value: 0.015751952672210624
```

6. Interpretation of Results

The **very small p-values (<0.001)** for Chi-square tests indicate strong evidence against the null hypothesis.

This means:

- CMD prevalence is **significantly associated with gender**
- CMD prevalence **varies across age groups**

The **t-test result (p = 0.0158)** shows:

- A statistically significant difference between male and female CMD rates

7. Conclusion

All null hypotheses are rejected.

This indicates that:

- Gender significantly affects CMD prevalence
- Age significantly affects CMD prevalence
- CMD rates differ significantly between males and females

These findings highlight the importance of demographic factors in understanding mental health patterns.

8. Access to CSV file

The full hypothesis testing results are available in the CSV file:
Click the link below to access the file.

[CSV File](#)